

Problem 1:

Given a sequence of integers $a_1, a_2, \dots, a_n$. Find a contiguous subarray of exactly k consecutive elements such that the GCD of those k elements is maximized.

Input:

- Line 1: $n, k (2 \leq k \leq n \leq 10^3)$
- Next line: $a_1, a_2, \dots, a_n (1 \leq a_i \leq 10^6)$.

Output: The maximum GCD of k consecutive elements.

Example:

INPUT	OUTPUT
10 3 2 6 4 3 18 12 24 8 7 5	6

Problem 2:

Let $g(n)$ be the number of pairs (p, q) such that $p+q=2 \times n$ and p, q are prime. Two pairs are considered different if one of the two numbers belongs to one pair but not the other. Given integer n . Compute the value of the expression: $f(n)=g(2)+g(3)+\dots+g(n)$.

Input: Contains an integer $n (3 \leq n \leq 10^3)$.

Output: A single integer — the value of $f(n)$.

Example:

INPUT	OUTPUT
9	12

1) Let

- $g[i]$: The count of unique prime pairs (p, q) such that $p + q = 2i$ and $p \leq q$
$$g[i] = \#\{(p, q) \mid p + q = 2i, p, q \text{ are prime}\}$$
- $f[i]$: The cumulative number of valid prime pairs from 2 up to i .

$$f[i] = f[i - 1] + g[i]$$

2) To calculate $g[i]$, iterate through all primes $p \leq i$, and check whether $q = 2i - p$ is also prime (use Sieve of Eratosthenes – sàng nguyên tố).

$$f[n] = \text{sum}(g[i]) \text{ with } i \text{ in } [2, n]$$

3) Base case: $f[1] = 0, g[2] = 1$

4) Result: $f[n]$

Complexity: $O(N \log \log N)$ with $N = 2n$ (Sieve) + $O(n^2)$ (counting) $\Rightarrow O(n^2)$

Problem 3:

Given two integer sequences $A = (a_1, a_2, \dots, a_m)$ and $B = (b_1, b_2, \dots, b_n)$, find an element a_i in sequence A and an element b_j in sequence B such that $|a_i + b_j|$ is minimized ($1 \leq i \leq m; 1 \leq j \leq n$).

Input:

- Line 1 contains two positive integers $m, n \leq 10^5$
- Line 2 contains m integers $a_1, a_2, \dots, a_m$ (for all i : $|a_i| < 2 \times 10^9$)
- Line 3 contains n integers $b_1, b_2, \dots, b_n$ (for all j : $|b_j| < 2 \times 10^9$)

Output: Output the two indices i and j of the corresponding elements.

Example:

INPUT	OUTPUT
4 5 1 8 2 9 -5 -6 3 -7 -4	2 4

Explanation: $|a_2 + b_4| = |8 + (-7)| = 1$

Problem 4 (not DP → merge sort):

Given n and a sequence of n numbers $a_1, a_2, \dots, a_n$. An inversion is defined as a pair of elements a_i and a_j in the sequence satisfying $i < j$ and $a_i > a_j$.

Task: Count the number of inversions in the sequence.

Input:

- The first line contains an integer n ($1 \leq n \leq 5 \times 10^4$).
- The second line contains n integers $a_1, a_2, \dots, a_n$ ($-2^{31} \leq a_i \leq 2^{31}$).

Output: A single integer — the number of inversions in the sequence.

Example:

INPUT	OUTPUT
6 1 2 4 3 5 1	5

Problem 5:

Given string A and string B consisting only of lowercase letters. String B is said to appear at position i of string A if:

$$A_i = B_1, A_{i+1} = B_2, \dots, A_{i+\text{len}(B)-1} = B_{\text{len}(B)}$$

Task: Find all positions where string B appears in A .

Input

- Line 1: string A .
 - Line 2: string B .
- The lengths of A, B are at most 10^6 .

Output Print all found positions on one line in increasing order. If B does not appear in A leave the output empty.

Example:

INPUT	OUTPUT
aaaaa aa	1 2 3 4

Problem 6 (not DP → Data Structure – Priority Queue, Min Heap):

Given k sorted sequences of integers (each sequence is non-decreasing). Elements may repeat.

Task: Merge all k sequences into a single non-decreasing sorted list.

Input

- The first line contains integer k — the number of sequences. ($1 \leq k \leq 10^4$)
- k The next k lines each contain one sorted sequence; each line has at most 500 integers.
- The total number of elements across all k sequences does not exceed 10^4 .

Output A single line containing all elements merged in non-decreasing order.

Example:

INPUT	OUTPUT
3 1 4 5 1 3 4 2 6	1 1 2 3 4 4 5 6

- 1) We use a min-heap (priority queue).

Each heap element stores:

- value
- which sequence it came from
- its position inside that sequence

Algorithm:

- Insert the first element of every sequence into the min-heap.
- While heap is not empty:
 - Extract the smallest element.
 - Add it to the output.
 - Insert the next element from the same sequence (if it exists).

- 2) (maybe skip)

- 3) We prove that the algorithm outputs all elements in non-decreasing order.

At every step:

- the heap contains the smallest unprocessed element from each sequence
- because each sequence is already sorted, any future element in a sequence is greater than or equal to its current heap element

⇒ The minimum element in the heap is the smallest remaining element among all sequences

After extraction: we insert the next element from the same sequence, maintaining the invariant

⇒ Every extracted element is the globally smallest remaining element, the produced output is sorted, every element is eventually processed exactly once

Hence the algorithm is correct.

- 4) Each heap operation costs: $O(\log k)$

Therefore total running time is: $O(N \log k)$

Space complexity: $O(k)$

Problem 7 (LeetCode problem):

Given a sequence of n non-negative integers representing the elevation map of a terrain. After heavy rain, water is trapped (see illustration below). Water flows to adjacent lower positions if the current level exceeds the neighbor. Both ends of the terrain border the sea at level 0.

Task: Compute the total amount of water trapped after the rain.

Input

- The first line contains a positive integer n — the width of the terrain. ($0 \leq n \leq 3 \times 10^4$).
- The second line contains n non-negative integers h_i — the elevation at position i of the terrain. ($0 \leq a_i \leq 10^5$)



Output A single integer — the total amount of trapped water.

Example:

INPUT												OUTPUT	
12												6	
0	1	0	2	1	0	1	3	2	1	2	1		

Problem 8: (NQUEENS PROBLEM - Backtrack)

Given a chessboard of size $n \times n$. Find all ways to place n queens on the board such that no two queens attack each other (queens move along rows, columns, and diagonals). Below are 2 example placements of 4 queens on a 4×4 .

Input Contains integer n — the board side length and number of queens. ($1 \leq n \leq 9$)

Output

- The first line contains integer m — the number of distinct valid placements.
- The following lines contain matrices representing the board configurations, separated by blank lines.
- Each matrix has n rows. Each row contains n characters:
 - Character '.' represents an empty cell
 - Character 'Q' represents a cell containing a queen

Example:

INPUT	OUTPUT
4	2 .Q.. ...Q Q... ..Q. ..Q. Q... ...Q .Q..

Problem 9:

Given two strings s and p consisting of lowercase Latin letters ‘a’–‘z’. In pattern string p there may also be the special characters ‘*’ and ‘.’ with the following meanings:

- The character ‘*’ means the preceding character may be repeated zero or more times. For example, “abbbb” can be matched by “ab*”, and “a” can also be matched by “ab*”.
- The character ‘.’ matches exactly one arbitrary character. For example, “abc” can be matched by “a.c” or “ab.”.

Task: Determine whether string s can be fully matched by pattern p or not.

Input

- The first line contains string s . ($\text{len}(s) \leq 20$).
- The second line contains pattern p . ($\text{len}(p) \leq 30$).

Output Output “true” if string s can be fully matched by pattern p , and “false” otherwise.

Example:

INPUT	OUTPUT
mississippi mis*is*p*.	false
aab c*a*b	true

Problem 10:

Given a binary matrix of size $m \times n$ where each cell contains 0 or 1.

Task: Find the largest square submatrix containing only 0s or only 1s.

Input

- The first line contains two positive integers m and n — the number of rows and columns, respectively. ($1 \leq m, n \leq 100$).
- The next m lines each contain n space-separated 0s and 1s.

Output

- The side length of the largest square containing only 0s or only 1s.

Example:

INPUT	OUTPUT
4 5 1 0 1 0 0 1 0 1 1 1 1 1 1 1 1 1 0 0 1 0	2

- 1) Let $dp[i][j]$ be the side length of the largest square submatrix consisting entirely of the same value (all 0s or all 1s) whose bottom-right corner is at cell (i,j) .
- 2) If $a[i][j] = 1 \rightarrow dp[i][j] = 1 + \min(dp[i-1][j], dp[i][j-1], dp[i-1][j-1])$
else $dp[i][j] = 0$
- 3) Base cases: First row & first column $dp[i][0] = 1$, $dp[0][j] = 1$
- 4) Time & Space Complexity: $O(nm)$
Result: $\max(dp[i][j])$

Problem 11:

Consider k data clusters, each of which can be assigned to one of 2 groups: Group 1 or Group 2. Cluster i has weights (a_i, b_i) : if cluster i is assigned to Group 1 the weight gained is a_i ; otherwise the weight is b_i .

We need to select n clusters into Group 1 and m clusters into Group 2, with each cluster assigned to at most one group, so into $n+m \leq k$.

Task: Find an assignment of clusters to the two groups such that the total weight of all selected clusters is maximized.

Input:

- First line: k, n, m ($1 \leq k \leq 10^3; n + m \leq k$)
- Line i of the k following lines: a_i, b_i ($1 \leq a_i, b_i \leq 10^5$)

Output: A single integer — the maximum total weight.

INPUT	OUTPUT	Explanation
4 2 1 4 9 3 5 7 2 5 5	21	4 9 3 5 7 2 5 5

Problem 12:

Construct a square with integer area equal to a given positive integer S using integer coordinates in the range from -10^9 to 10^9

Input: A positive integer $S (S \leq 10^3)$.

Output: The 4 vertices of the square listed in clockwise or counterclockwise order. If no solution exists, output Impossible.

INPUT	OUTPUT
5	1 2 2 4 4 3 3 1
3	Impossible

Theorem: An integer S can be represented as: $S = a^2 + b^2$

iff every prime congruent to 3 (mod 4) appears with even exponent in the factorization of S .

Output: A square with vertices $(0,0)$, (a,b) , $(a-b, a+b)$, $(-b,a)$ has area $a^2 + b^2$

How to find a, b:

- Iterate a from $0 \rightarrow \text{floor}(\sqrt{S})$.
- For each a , calculate $b^2 = S - a^2$. If b^2 is a perfect square, you have found your coordinates.

Time complexity: $O(\sqrt{S})$

Problem 13:

Given n ($n \leq 10^3$) points with integer coordinates such that no two coincide and no three are collinear (n is even).

Task: Select 2 of the n points such that the line through them splits the remaining points into two equal halves.

Input:

- First line: n ($2 \leq n \leq 10^3$)
- Line i of the n following lines: x_i, y_i ($|x_i|, |y_i| \leq 10^6$) — coordinates of point i .

Output: Two integers — the indices of the two selected points.

Example:

INPUT	OUTPUT
6 3 5 1 3 3 1 6 1 8 3 6 5	2 5

Problem 14:

Given a convex polygon with n vertices with integer coordinates x_i, y_i . The vertices are listed counterclockwise and no three are collinear. Find a diagonal connecting 2 vertices that splits the polygon into 2 parts with minimum area difference.

Input

- First line: $n (n \leq 10^3)$
- Line i of the n following lines: $x_i, y_i (|x_i|, |y_i| \leq 10^9)$.

Output: Two integers — the indices of the two chosen vertices.

Example:

INPUT	OUTPUT
4 0 2 0 0 2 0 3 3	2 4

Shoelace Formula (for polygon Area with (counter)clockwise order)

$$\text{Area} = \frac{1}{2} \left| \sum_{i=1}^n (x_i y_{i+1} - x_{i+1} y_i) \right|$$

Algorithm: So for each i , we can:

- use a pointer j
- **expand j while $\text{area} \leq \text{Area}/2$**

Then check the best split near half.

Prove: Because the polygon is convex, as j moves counterclockwise, the area of triangle-based region increases monotonically.

Complexity: $O(n)$ (sum of each i + area)

Problem 15:

Given two circles on a plane with equal radius R and centers at (x_1, y_1) and (x_2, y_2) . Compute the area of the union of the two circles. All center coordinates and the radius are integers.

Input: Contains integers x_1, y_1, x_2, y_2, R ($1 \leq x_1, y_1, x_2, y_2, R \leq 100$)

Output: A single real number — the area of the union, to exactly 3 decimal places.

Example:

INPUT	OUTPUT
1 2 3 4 2	22.850

Problem 16:

Given a tree with n vertices. The distance between two vertices u, v is the number of edges on the path from u to v . The width of the tree is the sum of distances over all pairs $u, v (u < v)$. Determine the width of the tree.

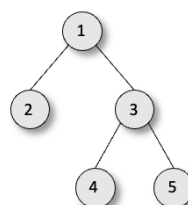
Input

- First line: $n (n \leq 10^3)$.
- $n - 1$ following lines: u_i, v_i — an edge of the tree.

Output: A single integer — the width of the tree.

Example:

INPUT	OUTPUT
5 1 2 1 3 3 4 3 5	18



(Contribution to the Sum Technique)

For any given edge in a tree, removing that edge splits the tree into two disjoint connected components.

1. Let the total number of vertices be n .
2. Suppose removing a specific edge e creates two components with k vertices and $n-k$ vertices, respectively.
3. Every path that starts at a vertex in the first component and ends at a vertex in the second component **must** pass through edge e .
4. How many such paths are there? Exactly $k \cdot (n-k)$.
5. Because every path crossing the edge adds exactly 1 to the total distance sum, edge e contributes exactly $k \cdot (n-k)$ to the total width of the tree.

The total width is simply the sum of these contributions for all $n-1$ edges in the tree

Problem 17:

Given a positive integer n . Find the smallest positive integer m with at most 9 digits satisfying

- m it contains only the digits $\{0, 1\}$
- m it is a multiple of n .

Input: A positive integer n ($n \leq 100$)

Output: A single integer — m

Example:

INPUT	OUTPUT
6	1110

Problem 18:

Given a sequence $a_1, a_2, \dots, a_n$ which is a permutation of integers from 1 to n . The following operation is allowed: choose a position i ($2 \leq i \leq n$) and reverse the prefix $a_1, a_2, \dots, a_i$, leaving the rest unchanged.

Task: Find the minimum number of operations to sort the sequence in non-decreasing order.

Input:

- First line: n ($1 \leq n \leq 8$)
- Second line: $a_1, a_2, \dots, a_n$

Output: A single integer — the minimum number of operations.

Example:

INPUT	OUTPUT	Explanation
5 5 2 3 4 1	4	5 2 3 <u>4</u> 1 4 3 <u>2</u> 5 1 2 3 4 <u>5</u> 1 5 4 3 2 <u>1</u> 1 2 3 4 5

Problem 19:

Given n consecutive segments; segment i has height h_i . The height difference between any two consecutive segments is at most 1. We want to excavate as deep as possible and plan to spend T coins. Reducing a segment's height by 1 unit costs 1 coin.

Task: Given the heights $h_1, h_2, \dots, h_n$ and budget T . Determine the minimum achievable height (deepest excavation) using at most T coins, while keeping consecutive height differences at most 1.

Input

- The first line contains two positive integers n, T ($n \leq 10^5, T \leq 10^{18}$)
- The second line contains n non-negative integers $a_1, a_2, \dots, a_n$ ($0 \leq a_i \leq 10^9$)

Output

A single integer — the minimum achievable height.

Example:

INPUT	OUTPUT	Explanation				
4 3 1 1 1 1	-1	1				
		0				
		-1				
4 3 1 2 2 1	0	2				
		1				
		0				

Problem 20:

Consider all permutations of $(1, 2, \dots, n)$. Assume they are sorted lexicographically and numbered from 1 to $n!$

Task: Given a permutation $(p_1, p_2, \dots, p_n)$ find its rank x ; and conversely: given a rank y ($1 \leq y \leq n!$) find the permutation $(q_1, q_2, \dots, q_n)$ with rank y .

Example for $n = 3$: there are 6 permutations: $(1,2,3)$; $(1,3,2)$; $(2,1,3)$; $(2,3,1)$; $(3,1,2)$; $(3,2,1)$

Input:

- Line 1: Contains n numbers $p_1, p_2, \dots, p_n$ ($n \leq 20$)
- Line 2: Contains number y

Output:

- Line 1: Write x
- Line 2: Write n numbers $q_1, q_2, \dots, q_n$

Example:

INPUT	OUTPUT
2 1 3	3
4	2 3 1

Problem 21:

Output the country SSS, there are N cities and M railway routes connecting them. All route lengths are distinct and it is guaranteed that every pair of cities is connected (directly or indirectly).

The government will upgrade a set of routes to a new high-speed rail system so that every pair of cities remains connected through upgraded routes. The goal is an optimal plan that minimizes the total length of upgraded routes (i.e., a Minimum Spanning Tree).

To handle contingencies, the government also wants to identify K distinct spanning-tree plans. Two plans differ if at least one route appears in one but not the other. Find the K -th backup plan for safety and reliability.

Input:

- The first line contains 3 integers N, M, K
- The next M lines each contain 3 integers u, v, c (edge from u to v with length c)

Output: Output a single integer — the total route length of the K -th spanning-tree plan.

Example:

INPUT	OUTPUT	Explanation
5 5 2 1 2 2 1 3 4 2 4 3 2 5 1 3 5 5	11	Routes in plan K include: 1 2 2 5 2 4 3 5

Problem 22:

In a mystical land, there is a kingdom called Graphonia with N cities and M roads connecting them. The King of Graphonia wants to optimize the road network. He wants to know the number of vertex pairs that become disconnected if each road is removed one at a time.

Input:

- The first line contains 2 integers N, M
- M The next M lines each contain 2 integers $u\ v$ (edge from u to v)

Output:

- Consists of M lines, each containing the number of vertex pairs disconnected when that edge is removed

Example:

INPUT	OUTPUT
5 5	0
1 2	0
2 3	4 (vertex 4 cannot reach
3 4	vertices 1, 2, 3, 5)
3 5	0
5 1	0